

SIMATS ENGINEERING
Department of Computer Science and
Engineering ASSESSMENT TOOL2 - CONCEPT MAPPING

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	C01	Assessment:	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	Total Marks:	25
C01 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
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Department:		Date:	

Instruction to Students

1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

ANNEXURE A

1. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.
2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.
3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.

4. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show real-world scenarios where specific addressing modes are preferred.
5. Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system design.

Code of Conduct:

I Sakthi Priya J 192411110 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					

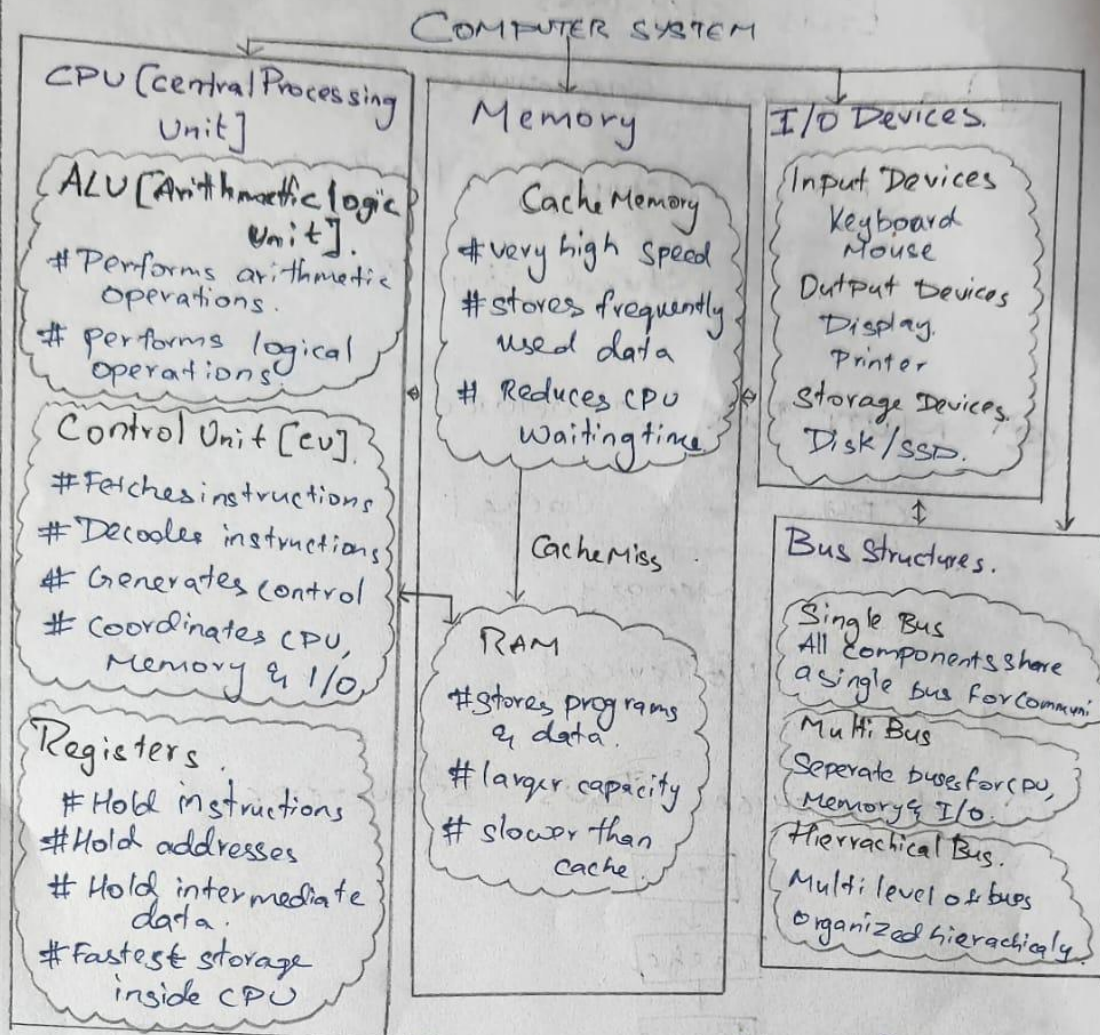
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

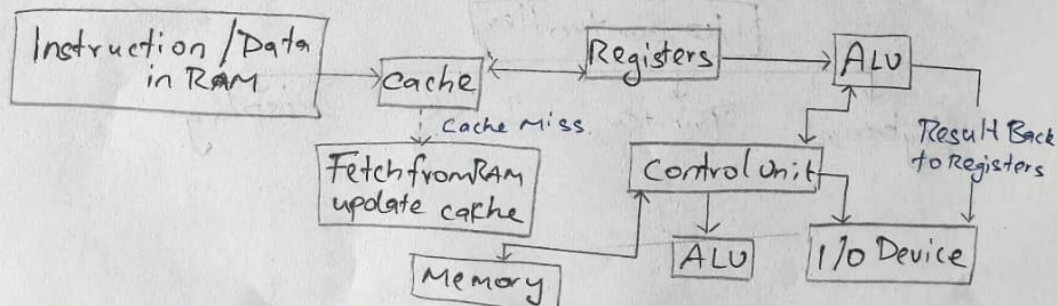
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Qno-1

CONCEPT MAP: CPU, MEMORY, I/O AND BUS STRUCTURES.



DATA & Control Signal Flow.



Qno-2.

INSTRUCTION EXECUTION CYCLE AND CPU PERFORMANCE

FACTORS AFFECTING EACH STAGE

Memory Access

- # Cache hit $\rightarrow$ Fast access
- # Cache miss $\rightarrow$ Main mem.
- # Affects Fetch & Execute

Instruction Complexity

- # Simple instructions take fewer cycles
- # Complex instructions take more cycles
- # Affects Decode and Execute time

Pipeline stalls / Hazards

- IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ MEM $\rightarrow$ WB
- # Data hazards
 - # Control hazards
 - # Structural hazards
 - # Causes stall cycles

INSTRUCTION EXECUTION CYCLE

FETCH STAGE

- # Program Counter holds address of next instruction
- # Instruction is read from memory.
- # Instruction loaded into instruction Register.

Time Contribution,
 T_{fetch}
[in clock cycles].

DECODE STAGE

- # Instruction decoded to determine op.
- # Control signals generated.
- # Source operands read from registers or memory.

Time Contribution,
 T_{decode}
[in clock cycles].

EXECUTE STAGE

- # Operation performed in ALU or functional unit
- # Memory access may occur.
- # Result written back to Register or memory.

Time contribution,
 T_{execute}
[in clock cycles]

BENCHMARKING: EVALUATING CPU PERFORMANCE

Select Benchmark Suite

Run on System

Collect Metrics

Analyze & compare

Qno-3.

8085 Vs 8086 Architecture Comparison.

8085 Architecture

8-bit Microprocessor

- # Total Instructions: 74.
- # 8-bit data operations
- # Simple instruction set
- # Most instructions 1 or 2 bytes.
- # NO MUL / DIV instruction.

REGISTER ORGANIZATION

- # 8-bit Registers
- # 16-bit Register
- # 16-bit PC
- # 16-bit SP.

Both are 8/16-bit microprocessors from Intel but differ in architecture features and capabilities.

KEY DIFFERENCE

- # 8085 is 8-bit, simpler
- # 8086 is 16-bit, more advanced.
- # 8086 has memory segmentation & more addressing modes

OVERALL Comparison.

Feature	8085	8086
Data Bus	8-bit	16-bit
Address	16-bit	20-bit
Instruction	74	111+
Registers	8-bit	16-bit
Address mode	6	9
Performance	Lower	Higher.

8086 Architecture.

16-bit Microprocessor [1978].

Instruction set.

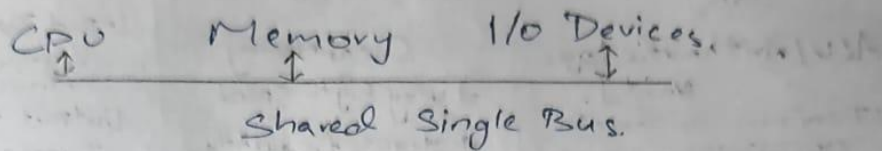
- # Total Instructions: 111+
- # 8-bit, 16-bit data operations
- # More powerful & complex.
- # Includes MUL, DIV, String, stack, flag, etc.
- # Instructions 1 to 6 bytes long.

ADDRESSING MODES.

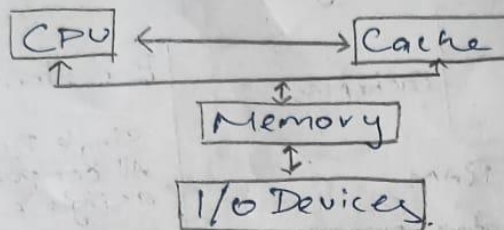
- # Immediate
- # Register
- # Direct
- # Register Indirect
- # Based
- # Indexed.
- # Based indexed.

Bus structures Comparison.

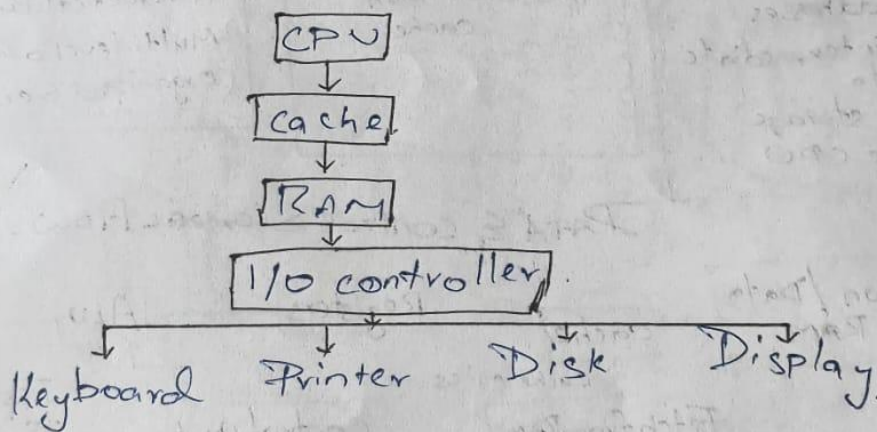
① SINGLE BUS STRUCTURE.



② MULTI BUS STRUCTURE



③ HIERARCHICAL BUS STRUCTURE



NUMBER SYSTEMS, INSTRUCTION REPRESENTATION, MEMORY STORAGE & DEBUGGING

NUMBER SYSTEMS

- Computer use only Binary internally
- Binary (Base-2): Uses 0, 1
- Hexadecimal (Base-16): Uses 0-9, A-F
- Hex is a shorthand form of binary

INSTRUCTION REPRESENTATION

- An instruction is a binary pattern
- It contains Opcode + operands
- The CPU interprets the opcode and performs the required operation using operands

MEMORY STORAGE

- Instructions are stored in memory as binary
- Memory addresses are shown in hexadecimal

CPU: FETCH-DECODE-EXECUTE

FETCH

DECODE

EXECUTE

DATA CONVERSION FLOW

Decimal → Binary → Hexadecimal → Stored in Memory.

Used during programming, assembling and debugging.